Tallinn University

School of Digital Technologies

DTLGM.DT Digital Learning Games

Designing an XR Drone Training Simulator

Master's Thesis

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Abstract

Automation is progressively displacing skilled technical labor, however, current robotic systems remain limited in replicating complex tasks, leaving a persistent need for human operators. At the same time, such skills are difficult to acquire, requiring prolonged training and the development of muscle memory. This creates a gap between the continued demand for skilled operators and the accessibility of training pathways.

This paper presents a design-based research study investigating Extended Reality (XR) as a medium for lowering the learning barrier and increasing engagement in complex skill acquisition, using First Person View (FPV) quadcopter piloting as a representative case domain.

This study is structured around three primary objectives: (1) the simulation of FPV quadcopter flight physics. (2) the design of a human-AI co-training framework (3) the intraction instruments for a high-fidelity XR prototyping. In particular PID and RL (Reinforcement Learning) are used in the co-training AI, and Geographic Information System (GIS) data integrated to support the environmental representation.

The contribution of this work is a structured design framework that links physical simulation, AI-assisted co-training, and XR-based interaction design, aimed at reducing the learning curve and improving accessibility to complex skill training.

Keywords: Extended Reality, Digital Twin, FPV Drone, Human-AI Co-training, PID Control, Reinforcement Learning

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Introduction

Robots are increasingly capable of replacing routine manual and cognitive tasks. However, there remains a strong demand for human cognitive and motor skills that machines cannot yet replicate. At the same time, this technological shift is occurring alongside a growing labor shortage, partly because many workers are unwilling to take jobs that are physically demanding and are expected to be replaced by automation in the future.

As a result, industries such as agriculture, forestry, and heavy-duty mobile machinery are already facing significant shortages of skilled workers (Machado et al., 2021). Moreover, career expectations are increasingly influenced by perceptions of automation. Research shows that when individuals believe a job is likely to be replaced by artificial intelligence, their willingness to pursue that career decreases significantly (Wang et al., 2026).

Despite the growing level of automation, current robotic systems still struggle with complex and dynamic environments. Therefore, skilled human operators remain essential. However, acquiring these skills is difficult and time-consuming, as it requires extensive repeated practice to develop muscle memory.

In this context, learning theory provides a useful framework. near transfer refers to applying skills in similar contexts, while far transfer refers to applying them in different contexts (Barnett & Ceci, 2002). For near transfer, repeated practice plays a critical role in building procedural knowledge and muscle memory.

Given these challenges, Extended Reality (XR) has emerged as a promising training approach. Previous studies have explored XR in domains such as medical procedures and aviation (Dutescu et al., 2025). Meanwhile, Unmanned Aerial Vehicles (UAVs) are widely used in construction, agriculture, logistics, and disaster management (Otto et al., 2018). Among these applications, First Person View (FPV) quadcopter piloting represents a particularly demanding skill that relies heavily on precise motor control and spatial awareness. Accordingly, this study uses FPV quadcopter piloting as a case to investigate XR-based training for muscle memory development.

However, existing drone simulators are primarily designed for entertainment rather than structured learning. Consequently, they often present steep learning curves and provide limited support for systematic skill acquisition. In addition, the lack of standardized controller configurations create further barriers (Ebeid et al., 2018), leading to user frustration even before effective training begins.

Taken together, low career expectations, high preparation effort, and early-stage frustration significantly reduce learner motivation. As a result, many beginners discontinue training before acquiring sufficient muscle memory.

To address these issues, the goal of this study is to design an XR-based training system that lowers the learning barrier for FPV quadcopter piloting. In particular, the proposed system aims to support muscle memory development through AI-assisted co-training and improved interaction design.

Accordingly, this study addresses the following research questions:

(1) How can FPV quadcopter flight physics be simulated?

(2) How can a human–AI co-training framework be designed to assist users?

(3) What interaction methods enable high-fidelity XR prototyping?

This study uses a design-based research approach. It applies agile development practices and iterative prototyping to build an XR FPV quadcopter simulator, including physics simulation, AI co-training (PID control and reinforcement learning), and interaction design.

Theoretical Background

Learning Theories in Simulation-Based XR

Experiential Learning

Active Learning

Game-Based Learning

Simulation Games

Game Design and Learning Frameworks

Game Design Framework

Learning Mechanics–Game Mechanics

Introduction to Quadrotor Systems

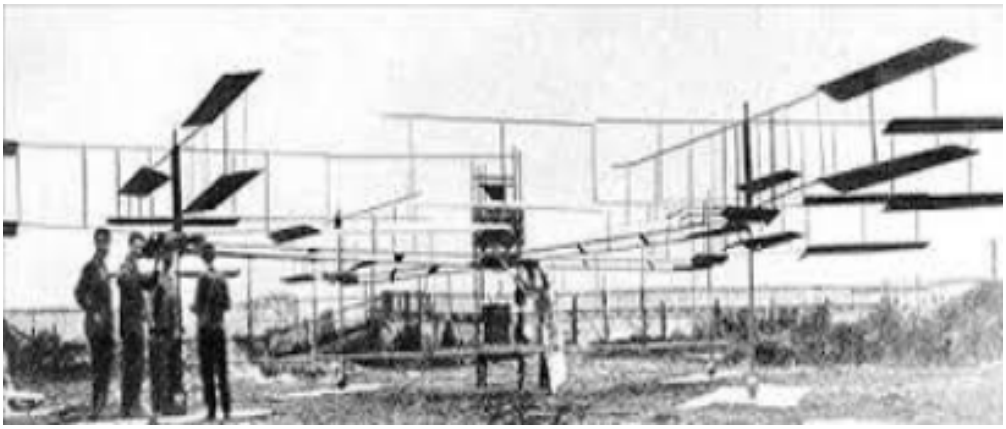
A quadcopter, also frequently referred to as a quadrotor, is a type of multirotor helicopter or rotary-wing Unmanned Aerial Vehicle (UAV) that utilizes four rotors to achieve flight (Luukkonen, 2011). These rotors are typically directed upwards and arranged in a square formation, with each rotor positioned at an equal distance from the center of mass. In this configuration, the four actuators are generally equally spaced by 90° from one another (Idrissi et al., 2022).

The origins of quadrotor technology date back to 1907, when the Breguet brothers constructed the Gyroplane No. 1 fig. 1 . While this marked one of the first manned helicopter flights, it was not a free flight; the machine lacked inherent stability and a proper control mechanism, requiring it to be tethered to the ground (Ghazbi et al., 2016).

The transition to modern UAVs occurred in the late 1990s. A pivotal moment was the 1999 release of the Draganflyer I fig. 2, the first camera-equipped quadrotor to achieve mass production. This shifted quadrotor technology from a laboratory curiosity to a viable commercial product (Karahan, 2025).

Since 2010, advancements in micro-electronics and control theory have expanded the use of quadrotors across various sectors:

Figure 1
Gyroplane No. 1 by the Breguet brothers



- Public Safety: Search and rescue, firefighting, and border surveillance.
- Agriculture: Precision seeding, chemical spraying, and crop monitoring.
- Commercial: Last-mile delivery and high-end cinematography.

Flight Dynamics and Principles of Operation

A quadcopter is an under-actuated system, meaning it must manage six degrees of freedom (6DOF) as shown in fig. 3, three translational and three rotational—using only four control inputs (the angular velocities of the rotors).

The fundamental principle of quadcopter stability relies on torque cancellation. To maintain a stable orientation, two motors rotate clockwise (CW) while the adjacent pair rotates counter-clockwise (CCW). This arrangement ensures that the net moment about the center of mass is zero; without this pair-wise opposition, the drone would rotate uncontrollably around its vertical axis. Quadcopters typically use either a "plus (+)" or "cross (x)" configuration, with the cross configuration being more common due to its superior stability (Idrissi et al., 2022).

The absolute linear position of the quadcopter is defined in the inertial frame axes ($[x, y, z]^T$). The attitude, i.e. the angular position, is defined in the inertial frame with three Euler angles ($[\phi, \theta, \psi]^T$), which represent pitch angle, roll angle and yaw angle(figure 3).

Every change in position or attitude is achieved by varying the individual angular

Figure 2
The Draganflyer I

velocities (ω_i) of the four rotors:

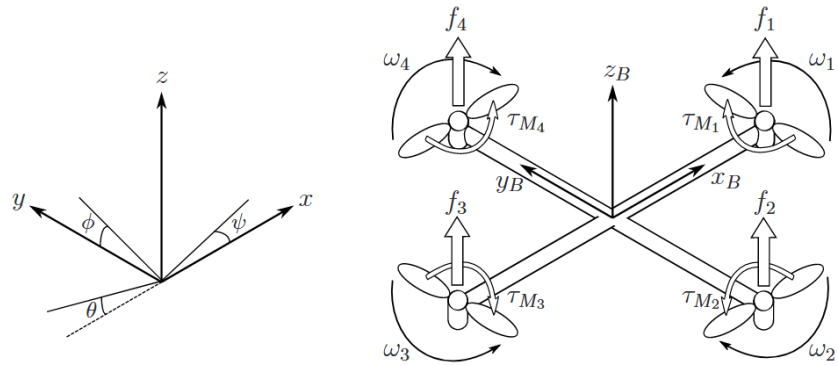
- Thrust (T): Simultaneously increasing all rotor speeds causes the vehicle to ascend. The force generated by rotor i is $f_i = k\omega_i^2$, where k is the lift constant. Total thrust is defined as $T = k\sum_{i=1}^{4} \omega_i^2$.

- Yaw (τ_ψ): Rotating the drone around the vertical axis is achieved by creating a torque imbalance between the CW and CCW pairs: $\tau_\psi = b(\omega_1^2 - \omega_2^2 + \omega_3^2 - \omega_4^2)$, where b is the drag constant.

- Roll (τ_ϕ) and Pitch (τ_θ): These are achieved by creating a speed differential between opposite rotors. For a distance l from the center of mass:

  - $\tau_\phi = lk(-\omega_2^2 + \omega_4^2)$
  - $\tau_\theta = lk(-\omega_1^2 + \omega_3^2)$

Figure 3
The inertial and body frames of a quadcopter



The quadcopter is modeled as a rigid body using Newton-Euler equations. The acceleration in the inertial frame $[x, y, z]^T$ is governed by gravity (g) and the orientation of the thrust vector:

$$\begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -g \end{bmatrix} + \frac{T}{m} \begin{bmatrix} \cos\psi \sin\theta \cos\phi + \sin\psi \sin\phi \\ \sin\psi \sin\theta \cos\phi - \cos\psi \sin\phi \\ \cos\theta \cos\phi \end{bmatrix}$$

In the body-fixed frame, the angular acceleration is influenced by the external torque (τ), inertia matrix (I), and gyroscopic effects:

$$\begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = \begin{bmatrix} (I_{yy} - I_{zz}) qr / I_{xx} \\ (I_{zz} - I_{xx}) pr / I_{yy} \\ (I_{xx} - I_{yy}) pq / I_{zz} \end{bmatrix} + \begin{bmatrix} \tau_\phi / I_{xx} \\ \tau_\theta / I_{yy} \\ \tau_\psi / I_{zz} \end{bmatrix}$$

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Discussion

Research Questions Revisited

XR supports training; AI provides guidance; system design partially validated.

Interpretation

XR enables immersion; AI provides feedback; combined supports learning.

Connection to Theory

Supports experiential and feedback-based learning.

Contribution

- XR-based training system
- AI-assisted co-training approach
- Integration of gamification and simulation

Limitations

No user study, limited RL, simplified tasks.

Future Work

User evaluation, more complex tasks, improved AI.

Summary

Goals

Develop XR training system with AI guidance.

Results

Functional system with immersive interaction.

Generalization

XR supports skill training.

Practical Use

Prototype training framework.

Future Study

Learning validation and system expansion.

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